

PSY304 stats demo

Shannon Dailey

10/29/2019

Descriptive Statistics

The dataset includes 248 participants total from 8 different cohorts (that is, 8 semesters of PSY304 students).

Age x gender breakdown of participants

Table 1: Age/gender breakdown of participants

	Girls	Boys	Total
3	49	41	90
4	55	60	115
5	15	25	40
Total	119	126	245

Summary of dependent variables

Table 2: Summary statistics of dependent variables

	Mean	Median	Minimum	Maximum
Age	3.8 (0.7)	4	3	5
Unoccupied	1.62 (2.79)	0	0	15
Solitary play	6.85 (6.58)	6	0	29
Onlooker	3.11 (3.92)	2	0	20
Parallel play	5.53 (6.01)	4	0	29
Associative play	6.79 (6.77)	5	0	30
Cooperative play	5.93 (7.56)	3	0	30
Social participation	3.98 (0.92)	4	2	5
Bossiness	1.51 (0.54)	1	1	3
Positive affect	2.82 (0.84)	3	1	4
Social awkwardness	1.51 (0.5)	2	1	2

Predominant play styles

A child's **predominant play style** is the type of play that child engaged in *the most* during the time sample.

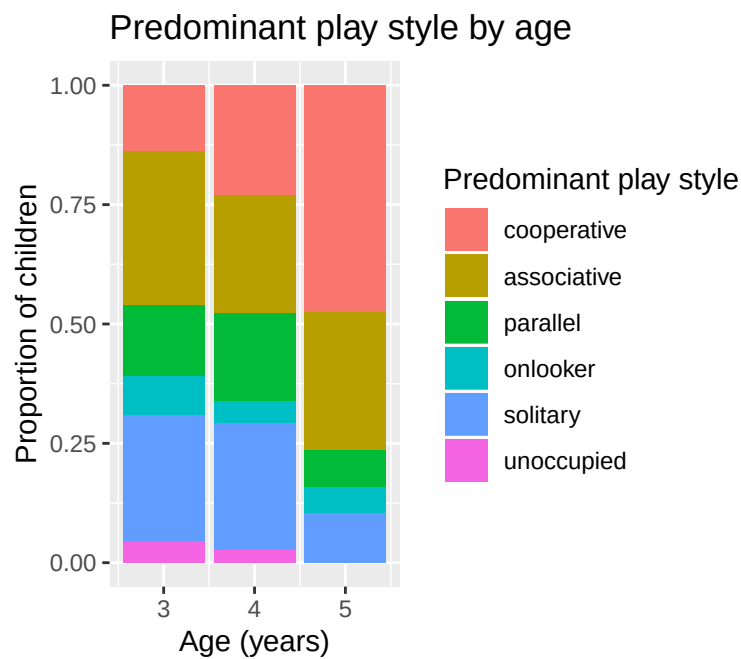
Table 3: Number of children with each predominant play style

Type of play	Count
cooperative	56
associative	66
parallel	37
onlooker	14
solitary	57
unoccupied	7

Predominant play style by age

Table 4: Predominant play style by age

	3	4	5
cooperative	12	25	18
associative	28	27	11
parallel	13	20	3
onlooker	7	5	2
solitary	23	29	4
unoccupied	4	3	0

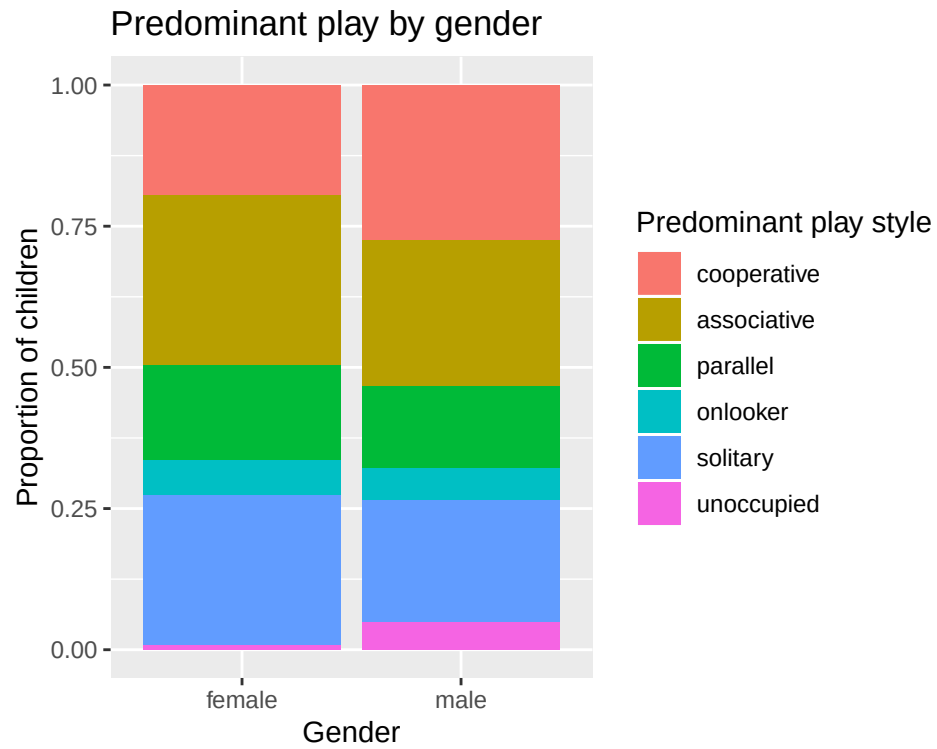


Chi-square test: $\chi^2(10) = 22.01$, $p = 0.02$

Predominant play style by gender

Table 5: Predominant play style by gender

	female	male
cooperative	22	34
associative	34	32
parallel	19	18
onlooker	7	7
solitary	30	27
unoccupied	1	6



Chi-square test: $\chi^2(5) = 5.89$, $p = 0.32$

Types of play

Types of play by age

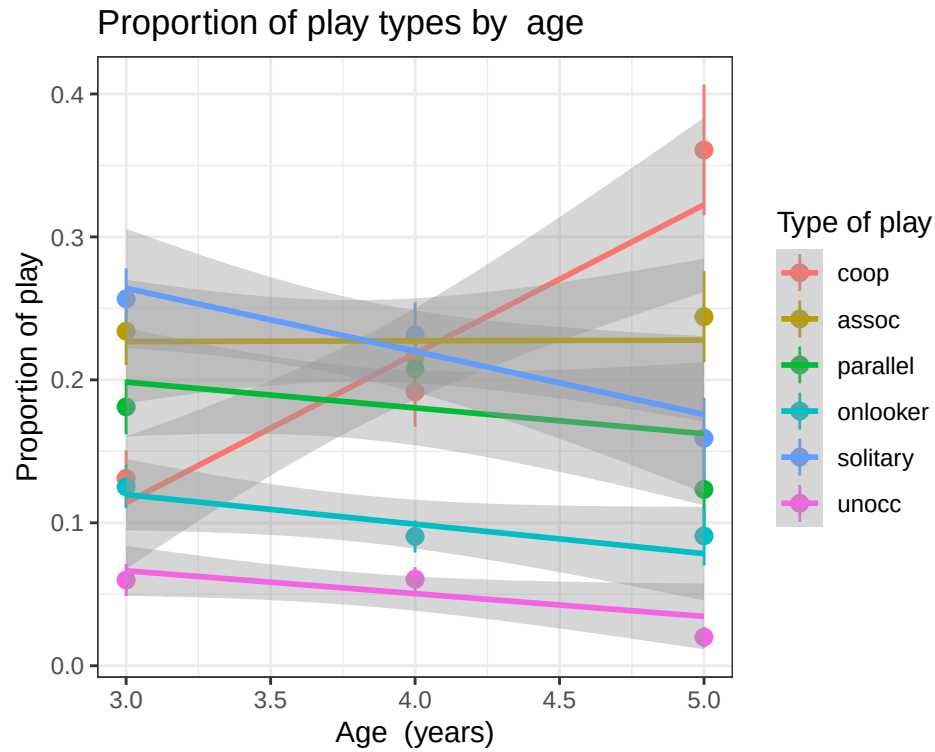


Table 6: Proportion of play types by child age

	3-year-olds		4-year-olds		5-year-olds	
	Mean	SD	Mean	SD	Mean	SD
Cooperative play	0.131	0.187	0.192	0.262	0.361	0.289
Associative play	0.234	0.225	0.216	0.237	0.244	0.201
Parallel play	0.181	0.182	0.208	0.214	0.123	0.186
Onlooker	0.125	0.143	0.090	0.120	0.091	0.130
Solitary play	0.257	0.203	0.232	0.242	0.159	0.179
Unoccupied	0.060	0.107	0.061	0.090	0.020	0.046

ANOVA: Proportion of cooperative play by age: $F(1, 243) = 22.01$, $MSE = 0.06$, $p < .001$

ANOVA: Proportion of solitary play by age: $F(1, 243) = 4.88$, $MSE = 0.05$, $p = .028$

Types of play by gender

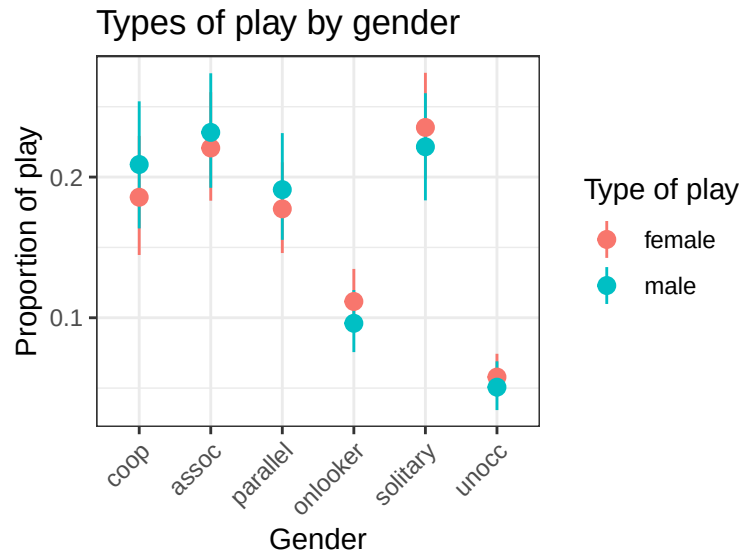


Table 7: Proportion of play types by child gender

	Girls		Boys	
	Mean	SD	Mean	SD
Cooperative play	0.186	0.247	0.209	0.257
Associative play	0.221	0.218	0.232	0.233
Parallel play	0.177	0.187	0.191	0.213
Onlooker	0.112	0.125	0.096	0.136
Solitary play	0.235	0.217	0.222	0.222
Unoccupied	0.058	0.086	0.051	0.099

T-test: Cooperative play by gender: $t(245.98) = -0.73, p = .469$

T-test: Solitary play by gender: $t(245.82) = 0.49, p = .623$